LINFO1115: Project – The Railway Network Analysis

Peter VAN ROY Augustin CRESPIN

Academic Year: 2025 – 2026

Context

How well does the railway network connect the different stations? Which stations play a central role in the infrastructure, and which ones are more isolated? Can we identify structural patterns such as tightly connected areas, bottlenecks, or critical links in the network? These are some of the questions you will investigate in this project.
To do so, you will work with a railway network dataset, where each station is represented as a node and each railway connection between two stations is represented as an edge weighted by the distance between them. The goal of this project is to analyze the structure of this network using graph-theoretic measures and to interpret the results in the context of railway infrastructure.

1 Guidelines

1.1 Dataset

For this project, there are two datasets. The first one (`total.csv`) represents the railway network for all of Europe, while the second one (`belgium.csv`) represents only the Belgian network. This smaller dataset is used only for Tasks 3 and 4.
Every entry of the dataset represents a connection between two stations. There are three columns in the dataset: `station_a`, `station_b`, and `distance_km`. A line of the dataset `(A, B, d)` therefore means that stations `A` and `B` are connected and that they are at a distance `d` of each other.
`A` and `B` are unique strings and represent the names of the station while the float `d` represents the distance in km.
To help you verify your answers to the different tasks, an open-source visualization of the dataset is available at the following link: `https://data-interop.era.europa.eu/map-explorer`.

1.2 In practice

The project is in Python. It is mandatory that the code compiles for it to be graded. The algorithms asked for this project must be implemented *by yourselves*: in particular, it is forbidden to use any library implementing them already. Of course, you may use any available tools for validation purposes.

This assignment must be completed by a **group of two students or alone**. If you do not manage to find a partner, please start by requesting a partnership via the Moodle forum. Then, **if and only if** you cannot find anyone else to work with, get in touch with us.

Unless specified otherwise, for each of the items below, we request you to explain your quantitative results. For each algorithm, indicate also the temporal complexity in your report. This complexity should be expressed using the number of vertices V and/or the number of edges E.

1.3 Deadline

This assignment is due on **May 3rd at 11:55 pm**. The code and the report analyzing the obtained results must be submitted on INGInious (`https://inginious.info.ucl.ac.be/course/LINFO1115`). A code template is available on Moodle with some details about what should be returned for each question.

2 Tasks

Task 0: 4 points (Easy)

This introductory task aims to verify that you can manipulate the dataset. To do so, we ask you to answer these very simple questions:

1. How many distinct nodes are there in the dataset?
2. How many edges?
3. Is the graph connected? (The visualization of the graph can be really helpful here!). Give the number of components.
4. Give the name of the nodes constituting the only path joining `Ahlbeck_Grenze` to `Peenemunde` (located on the Usedom island in Germany). What is the length of this path? (i.e. the rail distance in kilometers)
5. What is the length of the shortest path from `Portarlington_Junction` (located next to the Portarlington Station) to `Foyens_Junction` in Limerick, Ireland? To answer this question, no need to implement a shortest path algorithm, you can find all the possible paths (clue: there are 11 of them) and return the minimal length among them.

Task 1: 4 points (Easy)

In this first exercise, we are interested in knowing more about the general structure of the graph. Therefore, we ask you to provide:

1. The average degree of the nodes (average number of connections per node)
2. The degree distribution histogram of the graph (degree 10 and below).
3. The number of bridges in the graph

4. The number of local bridges in the graph.

How can you interpret those numbers in the context of railway infrastructure?

Task 2: 4 points (Intermediate)

First, give the clustering coefficient of the following stations:

- `Berlin_Westhafen`
- `Krakow_Gowny`
- `Amsterdam_Transformatorweg_Aansl.`
- `ROMA_TERMINI`

As a reminder, the clustering coefficient of a node A is defined as:

$$CC(A) = \frac{\text{number of pairs of A's connection that are connected}}{\text{maximum number of pairs of A's connections that are connected}}$$

We would like to analyze further how close the neighboring stations of a given station are to each other. For the following, consider that if stations A and B are close (less than a kilometer apart), the edge can be labeled as positive (+). If the stations are far (more than a kilometer apart), then the edge can be labeled as negative (-). Therefore, we ask you to

1. Compute the number of triangles contained in the graph.
2. Count the number of balanced and unbalanced triangles.
3. Compute the global clustering coefficient (GCC).

As a reminder, the GCC is defined as

$$GCC(G) = \frac{\text{number of closed triplets}}{\text{number of open triplets}}$$

where a triplet is a set of three distinct nodes. A triplet is said to be open if the three nodes are connected by two edges and closed if the three nodes are connected by three edges. A triangle contains three closed triplets, each centered on one of the three nodes. Interpret your results.

Task 3: 4 points (Advanced)

For this task, you must use the Belgian dataset

We ask you to compute the betweenness centrality which is used to indicate the importance of each node in a graph. For each node, you must compute the shortest path (in terms of **distance**) between each pair of nodes and count the number of times each node of the network is part of such shortest path. We ask you to give the name of the station with the highest betweenness centrality and the computed value.
Then, normalize this score by dividing by $\mathcal{C} = \frac{(V-1)(V-2)}{2}$

Task 4: 4 points (Advanced)

For this task, you must use the Belgian dataset

In order to improve the service, we would like to know how well connected the stations are. To do so, we ask you to compute the following score for each station:

$$score(n) = deg(n) \times \sum_{i \in Neighbors(n)} \frac{deg(i)}{d(n,i)} \times \sum_{j \in Neighbours(i) \setminus \{n\}} deg(j) \times d(i,j)$$

Give the name of the 5 stations with the highest score (and the corresponding score) and the 5 stations with the lowest score (and the corresponding score).

We also ask you to create a histogram graph of the distribution of the score and to interpret it.

We would like to improve the overall score of the network. To do so, you have the possibility to add 5 stations exactly in the middle of an existing edge. Rank all edges in the original graph independently based on the criterion below and give the five edges having the highest criterion. For simplicity reasons, you do not need to iteratively insert the new stations: all five new stations are evaluated on the same original graph.

$$\text{criterion}(u,v) = \begin{cases} 0 & \text{if } score(u) + score(v) = 0 \\ \frac{\text{Score gain from splitting edge}(u,v)}{score(u)+score(v)} \times d(u,v) \times 100 & \text{otherwise} \end{cases}$$

3 Deliverables

Your report should be **maximum 4 pages** long, should have the following structure, and should contain the following information:

Name and NOMA of the two students

- Task 0
  - Results: Indicate the number of distinct nodes, edges, components, the nodes in the path from `Ahlbeck_Grenze` to `Peenemunde` and its length, and the length of the shortest path from `Portarlington_Junction` to `Foyens_Junction`.
  - Complexity: Indicate the time complexity of your algorithms.
- Task 1
  - Result: Give the average degree of the nodes, the corresponding histogram, the number of bridges and the number of local bridges.
  - Interpretation: Interpret those results in the context of the railway network.
  - Complexity: Indicate the time complexity of your algorithm.
- Task 2

    - Result: Give the CC of the three stations, the number of total triangles, balanced triangles and unbalanced triangles, and the GCC.
    - Complexity: Indicate the time complexity of your algorithm.

- Task 3
    - Result: Give the name of the station with the highest betweenness centrality (and the corresponding score)
    - Graph: Show the distribution graph of the betweenness centrality. Interpret.
    - Complexity: Indicate the time complexity of your algorithm.

- Task 4
    - Result: Give the name of the five stations with the highest/lowest score (and corresponding scores), present the distribution graph and give the five edges where the added stations will be.
    - Graph: Show the distribution graph of the score. Interpret.
    - Complexity: Indicate the time complexity of your algorithm computing the score.

4 Deadline

The assignment is due by **Sunday, May 3rd 2026 at 11:55pm**. The code and report should be submitted on INGInious (`https://inginious.info.ucl.ac.be/course/LINFO1115`) by the same deadline. Only one student per group of two should submit on INGInious.